

Matteo Martinelli, Ph.D.

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PROFESSIONAL PROFILE

Management Engineer & Automation Specialist | 5+ Years in Industrial Automation and Digitalization

I manage end-to-end industrial research and innovation projects for automation and digitalization: from lead generation and stakeholder engagement to use-case design, prototype development and technical/scientific dissemination.

Skills: Industrial Digitalization | Stakeholder & Partner Management | Technical Training & Public Speaking

Tools: Java, Python, Javascript; SQL/NoSQL, Data Warehouses; Docker, Apache Airflow, GCP, Git/GitHub; data analysis in Python; ML&AI fundamentals, Simulations (AnyLogic, PuLP), CRM (Twenty, Salesforce), multi-OS environments (Windows, macOS, Linux).

PROFESSIONAL EXPERIENCES

Senior Automation Engineer - Post Doctoral Research – UNIMORE

Reggio Emilia, Italy

Project Management | Lead Generation | Account Management

11/2024 – Ongoing

- **Generated 40+ industrial leads** by participating in innovation events, national fairs, and university-oriented meetups.
- **Designed industrial use cases** in complex production and logistics environment, aligning with the research group's capabilities and expertise;
- **Prepared and delivered 3+ industrial research projects** and co-authored 10+ scientific articles;
- **Coordinated a team of research analysts**, delivering **2 prototypes**, **2 articles** and the reconstruction of an existing MicroFactory equipment;
- Researched **Industrial Digital Twins**, Cyber-Physical **Architectures**, system **augmentation**, **AI-relationships**, patterns of **specialization**, use cases investigation, industrial **design techniques**.

AI, Automation & Software Development (Professional Coach) – Freelance

Reggio Emilia, Italy

Professional Training | Live Coaching | Public speaking

11/2024 – Ongoing

- Delivered **530+ coaching hours** to **more than 180 students** from both **industry** and **specialization programs for professionals**;
- **Prepared 5+ courses** with theoretical and practical material with a special focus on **translating complex technical knowledge into understandable concepts**.
- Setup and delivery of proficiency tests, **evaluation** of participants, coordination with training institutions;
- Main topics: Programming logics and **design tools**, **Python**, **Java** and **Object Oriented Programming**, data analysis with Python, **Docker** and containerization, **Data Warehouses**, **Airflow** and ETL pipelines, A principles of **cloud computing**, **AI** and **ML** principles, **Git** and GitHub usage.

Industrial Automation Expert - UNIMORE

Reggio Emilia, Italy

Industrial Research | Scientific Writing | Public Speaking

12/2020 – 11/2024

- Researched **Digital Twins** and **explainable AI (xAI)** in **industrial scenarios**, exploring synergies between **Industrial AI and Digital Twin** technologies and contributing to industrial use cases and publications.

Industrial Continuous Improvement & Safety Specialist - Ognibene Power  Reggio Emilia, Italy
Lean Manufacturing | Stakeholder Engagement | Audit Presentation | Team Management 06/2019 – 12/2020

- **Monitored and improved industrial processes** in assigned areas, through planned projects, stakeholders engagement and lean tools (SMED, WO, EWO-SEWO, Micro-Areas, Tag Management and others);
- **Successfully completed the 2020 audit coordinating a team of 4 people**, guaranteeing production targets and safety standards and **saving a total of 128 h/year in setups**.

Internship - Industrial Project Lead - Dana Motion System  Reggio Emilia, Italy
Lean Manufacturing | Assembly line | Industrial AI 01/2019 – 06/2019

- **Re-designed an existing assembly line** for a family of mechanical reducers, **increasing productivity from 46% to 81% in 5 months**, on schedule;
- Developed an **AI-based Poka-Yoke Proof of Concept** to prevent frequent assembly errors, **achieving over 90% classification accuracy**; results published in a scientific journal.

EDUCATION & TRAINING

Ph.D. in Industrial Innovation Engineering  Reggio Emilia, Italy
University of Modena and Reggio Emilia (UNIMORE) 12/2020 - 05/2024

- Published **15+ scientific articles** in **international journals** and **conferences** and participated in **4+ international conferences and workshops**
- Organized 1 international workshop (CCNC - IDETWIN 2026, Jan 2026, Las Vegas), co-edited 1 international special issue on **Digital Twins** and Emerging Technologies for **Industrial and Cyber-Physical Systems**

Guest Researcher at Universität Stuttgart  Stuttgart, Germany
Hosted at Institut für Steuerungstechnik der Werkzeugmaschinen (ISW) 10/2022 - 04/2023

- Researched **Digital Twins** for **Industry 4.0** and industrial composability challenges in machine tool and automation contexts.

TAILOR Summer School 2022 – Universitat Politècnica de Catalunya (UPC)  Barcelona, Spain
Advanced Course on AI 06/2022

- Explored challenges in **explainable** and **trustworthy AI**; awarded for **best research presentation**.

Master's Degree – Management Engineering  Reggio Emilia, Italy
University of Modena and Reggio Emilia (UNIMORE) 2016 – 2019

- Thesis: “An assembly line re-design: the Dana Motion System case.

Erasmus+ Program – Management Engineering  Krakow, Poland
AGH-University of Science and Technology 02/2017 – 07/2017

Bachelor's Degree – Management Engineering  Reggio Emilia, Italy
University of Modena and Reggio Emilia (UNIMORE) 2011 – 2015

- Thesis: “Implementation of a reading and translation system of DXF files into two axis PLC instructions”, applied in the packaging industry.

ADDITIONAL SKILLS AND INTERESTS

- **Languages:** Italian (native language), English (C1, fluent), German (A2, basic).
- Interests: Cooking, technology tinkering and home-labbing, socializing